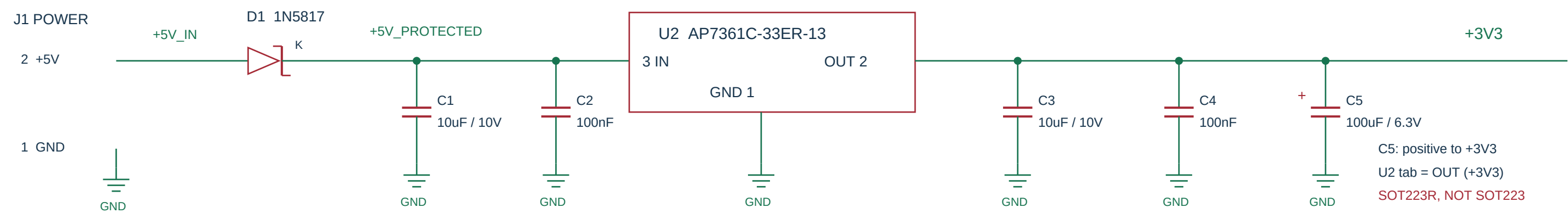


4VRS DISPLAY

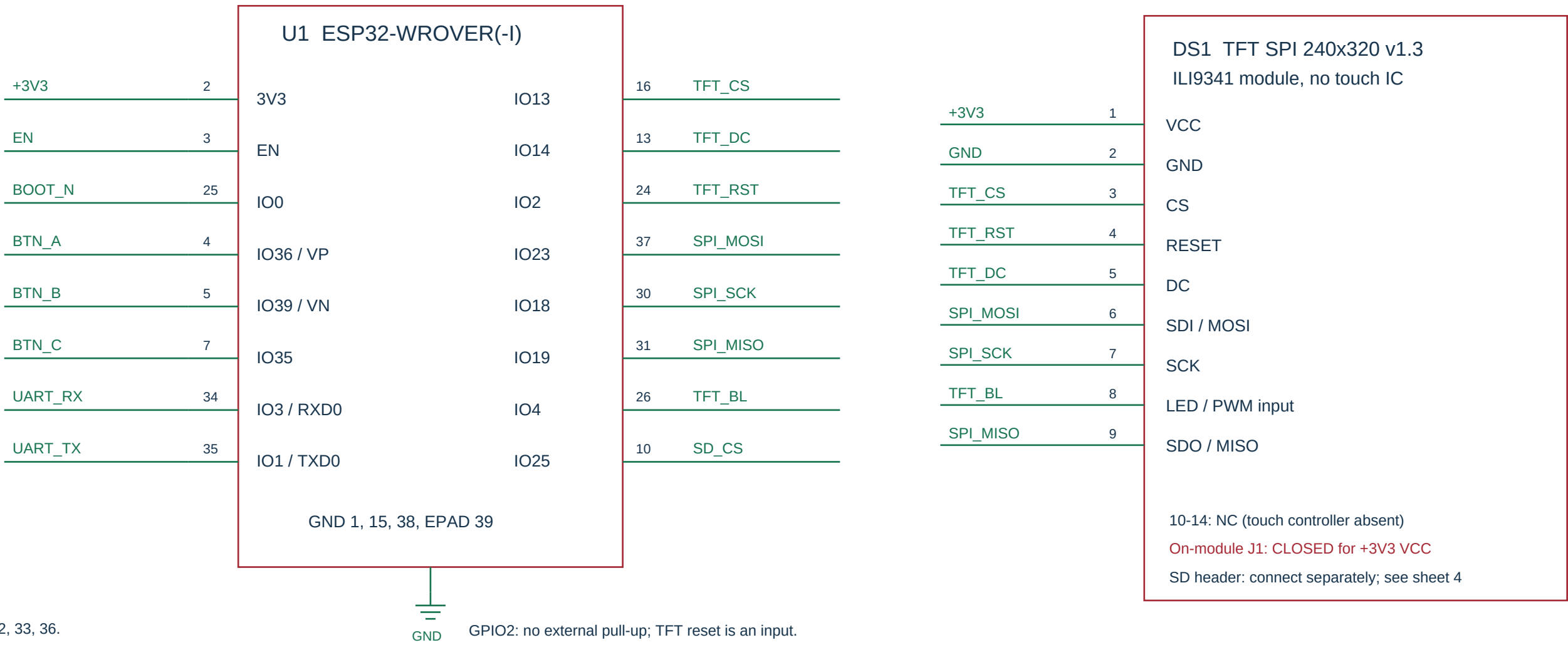
Power, controller and display

Electrical schematic | Rev A2 | 2026-10-05 | ESP32-WROVER + ILI9341 SPI 240x320

1. POWER - proposed lower-dropout supply; use a regulated 5 V / 1 A minimum source



2. SIGNALS - identical net names are electrically connected; numbers at U1 are module pad numbers



UNUSED / RESERVED U1 pads:
6, 8, 9, 11, 12, 14, 17-23, 27-29, 32, 33, 36.
Leave unconnected in this design.
Do not use flash / PSRAM pads as GPIO.

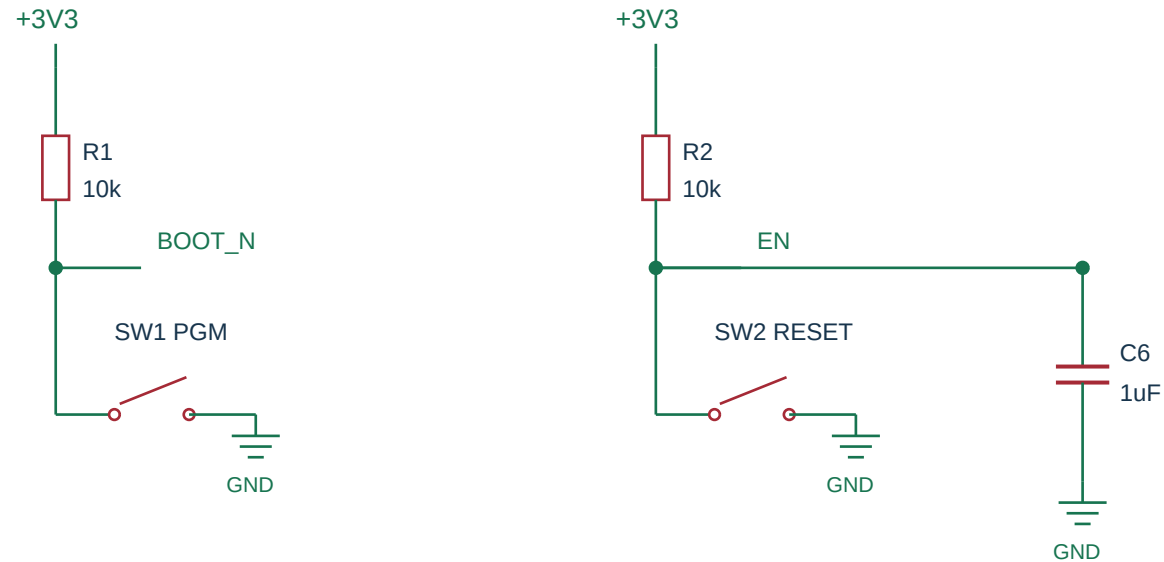
GPIO2: no external pull-up; TFT reset is an input.
GPIO12: leave unused; do not force a new boot strap.
WROVER(-I): attach the appropriate antenna before RF use.

4VRS DISPLAY

Reset, service connector and backlight

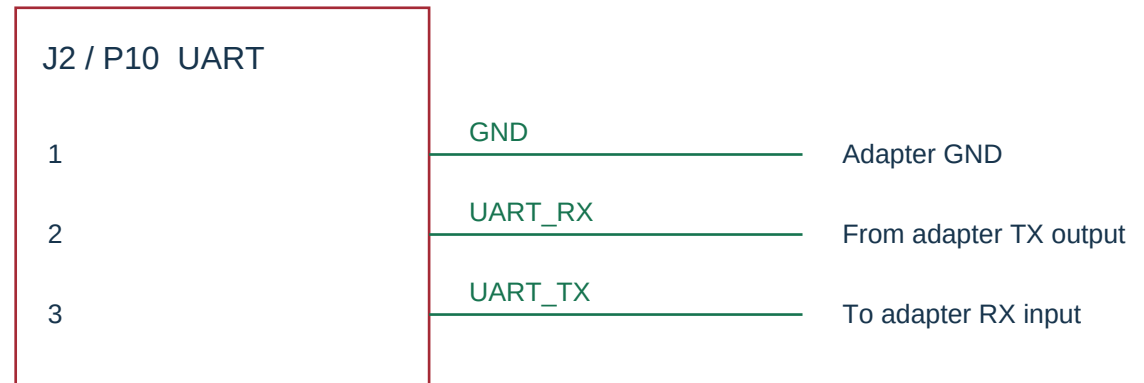
Electrical schematic | Rev A2 | 2026-10-05 | ESP32-WROVER + ILI9341 SPI 240x320

3. RESET / BOOT - added passive parts for defined startup



Hold PGM, pulse RESET, release PGM to enter the UART bootloader.
Normal start: RESET only. Do not short the EN pull-up resistor.

4. UART SERVICE - 3.3 V logic, manual reset



Power the device through J1; do not join two supply outputs.
Signal direction is authoritative: adapter connector labels may vary.
UART log: 115200 baud. GPIO1 = ESP TX; GPIO3 = ESP RX.

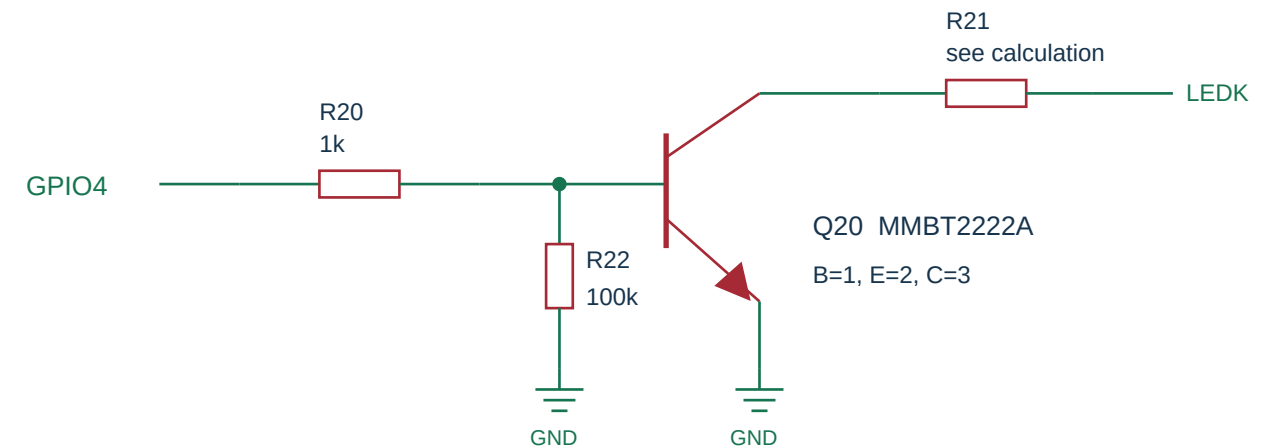
5. OUR MODULE - direct GPIO4 to display LED pin



Actual wiring: GPIO4 connects directly to DS1 pin 8 (LED).
No external transistor or series resistor in this connection.
Module internal backlight circuitry is not expanded here.
Firmware PWM: 5 kHz / 10 bit. HIGH enables backlight.



6. ALTERNATIVE ONLY - display with separate LEDA / LEDK



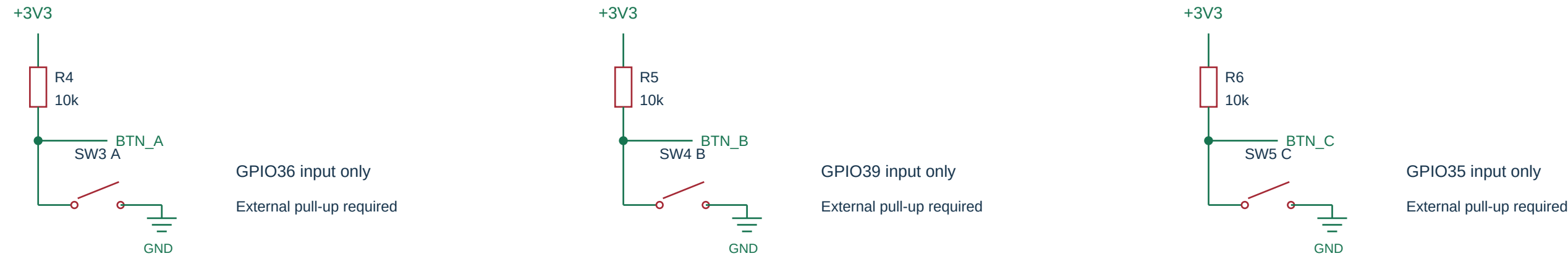
LEDA -> rated supply; LEDK -> R21 -> Q20 collector.
Example: one 3.0 V LED string, 3.3 V supply, 10 mA target.
 $R21 \geq (3.3 - 3.0 - 0.1) / 0.010 = 20 \text{ ohm}$; choose 22 ohm.
Recalculate for actual V_f , current and supply tolerances.
Parallel strings need individual limiting or a LED driver.
Not for our DS1 LED input. Do not connect both circuits.
R20 limits GPIO/base current; the original drawing omitted it.

4VRS DISPLAY

Assembly notes, options and component list

Electrical schematic | Rev A2 | 2026-10-05 | ESP32-WROVER + ILI9341 SPI 240x320

7. OPTIONAL BUTTONS - hardware provision; navigation is not implemented in firmware 1.0.0



8. MAIN ASSEMBLY BOM

U1	ESP32-WROVER(-I), PSRAM, module
U2	AP7361C-33ER-13, 3.3 V, SOT223R
D1	1N5817, series reverse-polarity diode
J1 / J2	2-pin power / 3-pin UART connector
DS1	TFT SPI ILI9341 240x320, LED logic input
C1 / C3	10uF X7R, 10 V; effective C >= 4.7uF
C2 / C4	100nF X7R, 10 V or higher
C5	100uF electrolytic, >= 6.3 V
C6	1uF X7R, >= 6.3 V
R1 / R2	10k, 1%, 0.1 W or higher
R3	100k, 1%, optional PWM pull-down
SW1 / SW2	Normally-open momentary PGM / RESET
R4-R6 / SW3-SW5	Optional: 10k / normally-open buttons

Q20 / R20-R22 belong only to the alternative, not the main BOM.

9. BUILD CONDITIONS AND REFERENCES

U2 is a proposed change from the original L1117-33 supply.

Check the exact SOT223R package: 1=GND, 2/tab=OUT, 3=IN.

Ordinary SOT223 AP7361C has a different pinout.

Place C1/C2 close to U2 IN; C3 close to U2 OUT.

Place C4/C5 close to U1 power pins with a short ground return.

Allow at least 500 mA for ESP32 plus LCD/backlight current.

LDO heat: $P=(V_{in}-3.3)*I$; 4.7 V at 0.5 A gives about 0.7 W.

Provide copper area and check temperature inside the enclosure.

For sustained high load, use a qualified 3.3 V buck supply.

DS1 VCC=3.3 V: close its supply jumper J1 (0R bypass).

If changing DS1 VCC to 5 V, open its J1 first.

Touch pins are NC; SD uses SPI + GPIO25 (sheet 4).

GPIO numbers are not ESP32 chip QFN pad numbers.

No ERC/SPICE or new PCB hardware validation has been run.

Confirm supply, startup and current on the assembled hardware.

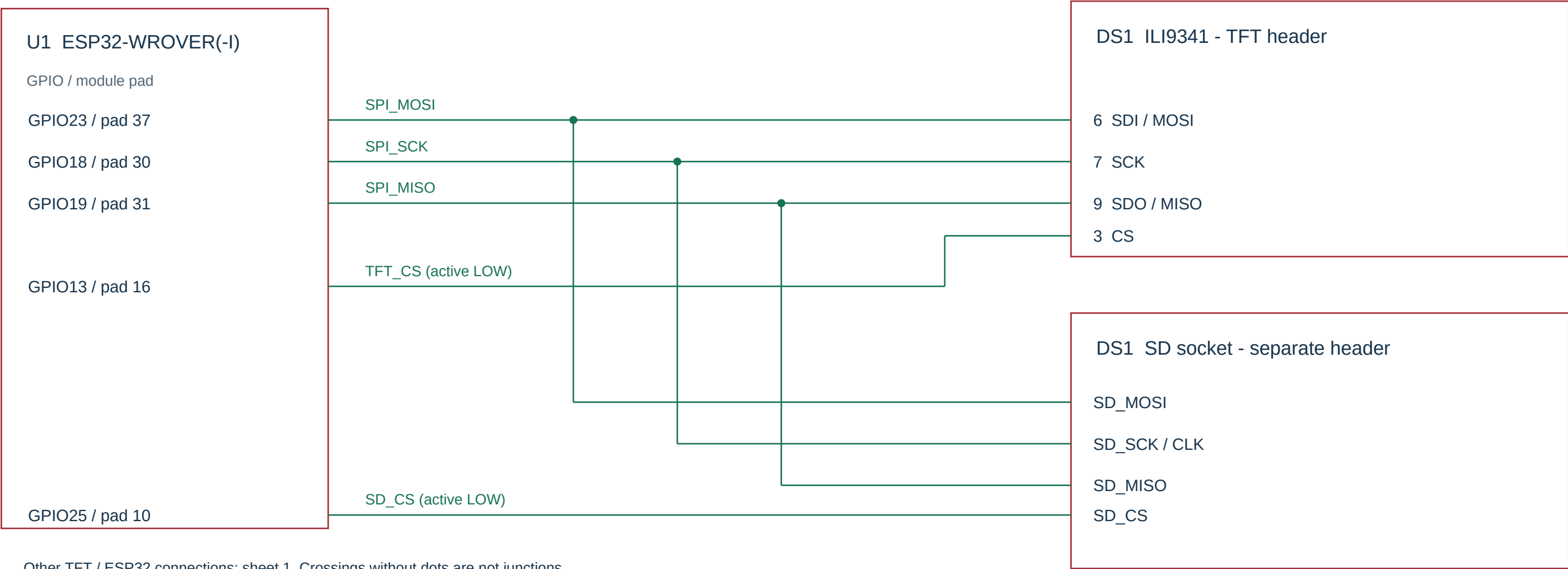
Manufacturer documentation (clickable in PDF):

LCDWIKI MSP2402: module schematic and manual

Espressif: WROVER module / hardware design guidelines

Diodes: AP7361C datasheet; original 1117: see TI datasheet

10. TFT + SD - common SPI signals, independent active-low chip selects



11. ASSEMBLY - existing SD socket on our LCDWIKI display module

Wire all FOUR SD header signals; the module does not internally connect them to the TFT SPI header.

SD header contacts are identified by their silkscreen names; no unverified header pin order is implied.

TFT_CS = GPIO13. SD_CS = GPIO25 (U1 module pad 10). Do not connect the two CS signals together.

SD socket power and ground are internal to DS1; retain the module VCC/GND connection from sheet 1.

All logic is 3.3 V. Keep SPI leads short. MISO is required for SD even without TFT readback diagnostics.

Firmware 1.0.0 uses a shared SPI bus, MISO pull-up and separate CS handling; SD starts at 1 MHz.

Use a FAT32 card; 16 GB microSD with a full-size SD adapter has been tested. No automatic formatting.

This sheet shows the existing module socket, not a wiring diagram for an arbitrary bare SD socket.